

How to Install and Run the IC Verify System

Step-by-Step Guide (Non-Coder Friendly)

Part 1: Create the Gemini API Key (AI Studio)

1. Go to **aistudio.google.com** in your browser.
2. Sign in with any Google account.
3. Click **"Get API key"** (usually top-left or in the left sidebar).
4. Click **"Create API key"**.
5. Choose **"Create API key in new project"** (or pick an existing Google Cloud project if you already have one).
6. Copy the key it generates — it's a long string of letters/numbers. Save it somewhere safe (like Notepad) for now, you'll paste it into the app in Part 4.

Part 2: Enable Google Drive & Google Sheets Access

The app reads IC files from Drive and writes results to a Sheet using a "service account" (a robot account, not your personal login).

1. Go to **console.cloud.google.com**.
2. At the top, make sure the **same project** is selected that you used in Part 1 (check the project name dropdown next to the Google Cloud logo).

Enable the two APIs:

3. In the search bar at top, type **"Google Drive API"** → click it → click **Enable**.
4. Search again for **"Google Sheets API"** → click it → click **Enable**.

Create the service account:

5. In the left menu, go to **IAM & Admin → Service Accounts**.
6. Click **"+ Create Service Account"** at the top.
7. Give it any name, e.g. `ic-verify-service`.
8. Click **Create and Continue** → skip the optional role/permission steps → click **Done**.

Generate its key file:

9. Click on the service account you just created (from the list).
10. Go to the **Keys** tab.
11. Click **Add Key → Create new key → JSON** → click **Create**.
12. A `.json` file downloads automatically to your computer. This is your `google_credentials.json` — keep it safe, don't share it publicly.

Copy the service account's email address — it's on the same service account page, and looks like:

`ic-verify-service@your-project-id.iam.gserviceaccount.com`

Share your Drive folder and Sheet with that email:

13. Open the Google Drive folder that has the IC files → click **Share** → paste the service account email → set access to **Editor** → click **Send/Share**.
14. Open the Google Sheet you're using as the database → click **Share** → paste the same email → **Editor** → **Share**.

Part 3: Install Node.js (one-time, only if not already installed)

1. Go to **nodejs.org**.
2. Download the **LTS** version for your OS.
3. Run the installer, click Next through the defaults.
4. Check it worked — open Command Prompt (Windows) or Terminal (Mac) and type:

```
node -v
```

You should see a version number like v20.x.x.

Part 4: Open and Set Up the Project in Antigravity

1. Download and install **Antigravity** from **antigravity.google** if it's not already installed.
2. Open Antigravity.
3. Click **Select Project** (or File → Open Folder) and choose your Geetha project folder.
4. Antigravity will load the project with its file explorer and a built-in terminal at the bottom.
5. In that built-in terminal, place the two files you got earlier:
 - Copy `google_credentials.json` (from Part 2, step 12) into the folder: `server/config/google_credentials.json`
 - In the project's root folder, find `.env.example`, duplicate it, and rename the copy to `.env`. Open it and fill in:

```
GEMINI_API_KEY=paste_your_key_from_part_1_here
PORT=5000
GOOGLE_APPLICATION_CREDENTIALS=server/config/google_credentials.json
```

Part 5: Install Dependencies & Run the App

In Antigravity's built-in terminal (or any terminal opened inside the Geetha folder), run these commands **in order**:

```
npm install
```

Wait for it to finish — this installs everything the app needs.

Then start both servers. Easiest is two terminal tabs/windows inside the same folder:

Terminal 1 (backend):

```
npm run server
```

Terminal 2 (frontend):

```
npm run dev
```

Or on Windows, just double-click `Start_App.bat` in the project folder — it runs both commands for you automatically and opens the browser.

Part 6: Confirm It's Working

1. Your browser should open to something like `http://localhost:5173` showing the Upload dashboard.
2. Upload a test Excel sheet, paste your Drive folder URL and Sheet URL, and run a verification on 1-2 students.
3. If you get a VERIFIED/REJECTED/REVIEW result back, everything is connected correctly.

If something fails: it's almost always one of — API key missing from `.env`, `google_credentials.json` in the wrong folder, or the Drive folder/Sheet not shared with the service account email (Part 2, step 14).